

Supplementary Information

Separating effect size from statistical evidence in network-proximity rankings under target-count asymmetry

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Supplementary Notes

Network type. STRING combined-confidence edges integrate multiple evidence channels (experiments, curated databases, co-expression, genomic context, text mining); the resulting graph is a *functional association network*, not a purely physical protein-protein binding network. We use “interactome” in this broad STRING association-network sense throughout.

Text-mining robustness. Because the DILI module is DisGeNET-derived (literature-based) and the STRING combined score includes a text-mining channel, some target-DILI edges could reflect literature co-mention rather than independent evidence. Following Huang et al. [1], we rebuilt the interactome after removing every edge supported *solely* by the STRING text-mining channel (all other evidence channels zero) and re-ran all three metrics; a from-scratch rebuild without this filter recovers all committed network edges (0 missing, with a small mapping superset), and the full-rebuild observed values match the committed values, so the comparison is mapping-consistent. Text-mining-only edges constitute 0.7% (≥ 900) to 2.4% (≥ 700) of the liver-expressed LCC. Removing them leaves the effect ordering intact: Hyperforin remains closer ($d_c = 1.30$ vs 1.67 at ≥ 900) and higher in per-target random-walk (0.114 vs 0.033) and expression-weighted (0.134 vs 0.050) influence, with the ordering preserved on all three metrics at both thresholds (all $p < 0.002$; per-metric values in Table S5, full output in `results/tables/string_textmining_sensitivity.csv`). This argues against literature text-mining circularity as the driver of the effect/evidence dissociation, consistent with Huang et al.’s finding that STRING’s disease-gene signal is robust to text-mining removal.

DILI-module provenance. The DILI gene set is derived from DisGeNET curated gene-disease associations (UMLS C0860207). It is a *curated association module*, not a validated causal gene set: some members are metabolism genes, pharmacogenes, or literature-associated biomarkers rather than established causal DILI drivers. This is relevant because four Hyperforin targets (NR1I2, CYP2C9, ABCB1, MMP2) are themselves members of this module; the direct-propagated decomposition (Table S9) quantifies the resulting direct overlap.

Null estimands. Two degree-matched permutation nulls are reported. The *fixed-disease null* randomises only the target set and holds the disease module at the 82 real DILI genes; it evaluates whether the observed target set is unusually close to the fixed DILI module and is the appropriate primary null for comparing two compounds against the same disease. The *two-sided null* (after Guney et al. [2]) additionally randomises a degree-matched disease set; it answers the population-level drug-disease question and is reported as a fidelity sensitivity (Table S13).

p-value convention. All permutation p-values are empirical, directional, and computed as $(r + 1)/(n + 1)$ [3]: one-sided for *smaller-than-null distance* (proximity) and *larger-than-null influence* (RWR/EWI). With $n = 1,000$ the floor is 9.99×10^{-4} . Z-scores are the primary standardized-evidence statistic; no two-tailed permutation p-value is used.

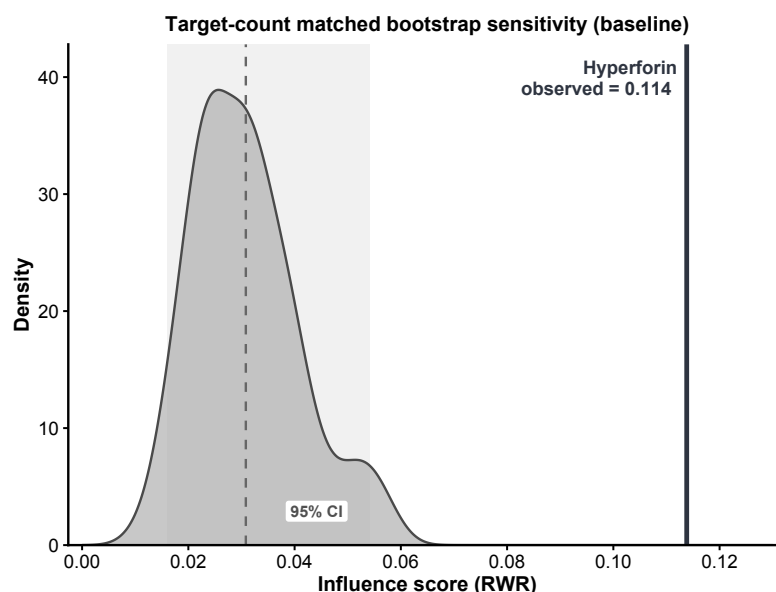


Figure S1. Bootstrap sensitivity (baseline). Distribution of RWR influence for random 10-target subsets of Quercetin; solid line, Hyperforin observed influence. This baseline control does not adjust for target–DILI overlap; the leakage-controlled analysis (Table 3, Fig. 6) supersedes the raw baseline ratio. STRING ≥ 900 .

Supplementary Figures

Supplementary Tables

All values above are regenerated by the committed pipeline and reviewer-evidence scripts; ratios, standardized scores, and separation values are computed from full-precision underlying data and may differ in the last displayed digit from the quotient of the rounded entries shown; cross-document numerical consistency is enforced by `verify_numbers.py`.

References

- [1] Justin K Huang, Daniel E Carlin, Michael Ku Yu, Wei Zhang, Jason F Kreisberg, Pablo Tamayo, and Trey Ideker. A systematic evaluation of molecular networks for discovery of disease genes. *Cell Systems*, 6(4): 484–495, 2018. doi: 10.1016/j.cels.2018.03.001.
- [2] Emre Guney, Jörg Menche, Marc Vidal, and Albert-László Barabási. Network-based in silico drug efficacy screening. *Nature Communications*, 7:10331, 2016. doi: 10.1038/ncomms10331.
- [3] Belinda Phipson and Gordon K Smyth. Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1):Article 39, 2010. doi: 10.2202/1544-6115.1585.
- [4] Jörg Menche, Amitabh Sharma, Maksim Kitsak, Susan Dina Ghiassian, Marc Vidal, Joseph Loscalzo, and Albert-László Barabási. Uncovering disease-disease relationships through the incomplete interactome. *Science*, 347(6224):1257601, 2015. doi: 10.1126/science.1257601.

Table S1. Target curation evidence. Hyperforin targets are literature-curated and span a direct nuclear receptor (PXR/NR1I2), PXR-induced downstream enzymes, transporters, and direct-inhibition targets; Quercetin targets are uniformly ChEMBL bioactivity-derived. This documents the evidentiary asymmetry whose downstream consequence is audited in Table S9.

Gene	Source	Evidence type	Direct	In DILI	Shared	Kept [†]
<i>Hyperforin (literature-curated)</i>						
NR1I2	literature	direct receptor agonist (PXR)	yes	yes	no	yes
CYP3A4	literature	PXR-induced enzyme	no	no	yes	no
CYP2C9	literature	enzyme (inhib./induction)	partial	yes	no	no
CYP2B6	literature	PXR-induced enzyme	no	no	no	no
ABCB1	literature	transporter (induced + inhib.)	yes	yes	no	no
ABCG2	literature	transporter modulation	partial	no	yes	no
ABCC2	literature	transporter modulation	partial	no	no	no
AKT1	literature	direct kinase inhibition	yes	no	yes	yes
MMP2	literature	secretion inhibition	partial	yes	yes	yes
MMP9	literature	secretion inhibition	partial	no	yes	yes
<i>Quercetin (ChEMBL bioactivity, ChEMBL159; 62 targets, full list Table S14)</i>						
in vitro binding/activity ($IC_{50}/K_i/EC_{50} \leq 10 \mu M$)			yes	MMP2 (1)	5	n/a

“Direct” classifies the target-compound interaction: *yes* = direct molecular target (receptor agonism or enzyme/kinase inhibition); *partial* = mixed or context-dependent direct engagement (e.g. both induction and inhibition); *no* = downstream induced effect rather than direct binding. [†]Kept = retained in the “CYP/transporter-out” leakage control (Table S9).

Table S2. Hyperforin targets in the liver-expressed LCC (STRING ≥ 900). Node degrees are recomputed in the analysis network (correcting earlier values taken from a larger network). TPM from GTEx v8 liver.

Gene	Protein	TPM	Degree	DILI link
NR1I2	PXR	43	4	direct (module member)
CYP3A4	CYP3A4	335	73	indirect
CYP2C9	CYP2C9	434	52	direct (module member)
CYP2B6	CYP2B6	125	34	indirect
AKT1	PKB	33	128	indirect
ABCB1	P-gp	7	1	direct (module member)
ABCC2	MRP2	60	1	indirect
ABCG2	BCRP	4	2	indirect
MMP2	MMP-2	5	20	direct (module member)
MMP9	MMP-9	1	25	indirect

Table S3. Permutation null parameters across metrics (STRING ≥ 900 ; $n = 1,000$ degree-matched permutations). The null standard deviation σ_{null} of the 62-target compound is approximately $2.5\times$ smaller than that of the 10-target compound for shortest-path proximity, random-walk influence, and expression-weighted influence, consistent with Law-of-Large-Numbers shrinkage ($\sqrt{62/10} = 2.49$) across metrics. This is why Z-score magnitude is not comparable across compounds of very different target count.

Metric	Compound	μ_{null}	σ_{null}	x_{obs}	Z
Shortest-path d_c	Hyperforin (10)	2.2086	0.2353	1.3000	-3.86
	Quercetin (62)	2.1749	0.0914	1.6774	-5.44
RWR influence	Hyperforin (10)	0.0141	0.0093	0.1138	10.77
	Quercetin (62)	0.0148	0.0038	0.0322	4.58
EWI influence	Hyperforin (10)	0.0196	0.0122	0.1330	9.28
	Quercetin (62)	0.0210	0.0049	0.0493	5.74
σ_{null} ratio (Hyperforin/Quercetin): shortest-path 2.57, RWR 2.45, EWI 2.47 (expected $\sqrt{62/10} = 2.49$).					

Table S4. Operating-regime benchmark: margin-conditional rank-reversal rates (liver LCC ≥ 900 , DILI module; probes pinned to the full compound-target row degree profile, with DILI-module genes excluded from the candidate pool; smaller set $|T_S| = 10$; 500,000 probe pairs per row, seed 42). Each entry is the fraction of pairs in which the smaller set is genuinely closer by a material raw margin ($\geq \delta_0$ hops) yet the larger set has the more negative Z-score. Reversals are rare and require a smaller set above the 90th percentile of probe-pair margins: a no-shrinkage counterfactual ($\sigma_L := \sigma_S$) gives 0% at every R , and unconditionally the Hyperforin/Quercetin directional pattern (smaller set closer *and* larger set stronger Z) occurs in only 6.5% of exact $R = 6.2$ pairs, collapsing to near-zero once a material margin is required. The null-SD exponent is -0.499 (95% CI $[-0.502, -0.495]$, $R^2 = 0.9999$) and similar across the DILI module and three size-matched pseudo-modules (DILI -0.495 ; three random size-matched pseudo-modules mean -0.498).

$R = T_L / T_S $	(T_S , T_L)	reversal, $\delta_0 \geq 0.3$ (%)	reversal, $\delta_0 \geq 0.5$ (%)
1.5	(10, 15)	0.00	0.00
2.0	(10, 20)	0.00	0.00
3.0	(10, 30)	0.00	0.00
4.0	(10, 40)	0.00	0.00
6.0	(10, 60)	0.10	0.00
6.2	(10, 62)	0.06	0.00
8.0	(10, 80)	0.39	0.00

Table S5. Text-mining robustness: metric values before and after removing text-mining-only STRING edges (liver-expressed LCC; following Huang et al. [1], every edge supported *solely* by the STRING text-mining channel was removed). “Committed” is the value in the main analysis; “no-TM” is after removal. The Hyperforin > Quercetin ordering (smaller d_c ; larger influence) is preserved on all three metrics at both network thresholds, and every no-TM directional permutation test gives $p < 0.002$, arguing against literature text-mining circularity as the driver. Source: `results/tables/string_text_mining_sensitivity.csv`.

Metric	STRING	Hyperforin		Quercetin		winner
		committed	no-TM	committed	no-TM	
d_c (smaller closer)	≥ 700	0.6000	0.6000	1.3387	1.3387	Hyp
	≥ 900	1.3000	1.3000	1.6774	1.6721	Hyp
RWR influence (larger)	≥ 700	0.1212	0.1243	0.0326	0.0335	Hyp
	≥ 900	0.1138	0.1144	0.0322	0.0329	Hyp
EWI influence (larger)	≥ 700	0.1449	0.1472	0.0511	0.0516	Hyp
	≥ 900	0.1330	0.1337	0.0493	0.0501	Hyp

Table S6. Threshold robustness (STRING ≥ 700 vs ≥ 900 liver LCC; $n = 1,000$, seed 42). The proximity-Z *ranking* reverses across thresholds — Hyperforin is the more “significant” at ≥ 700 , Quercetin at ≥ 900 — whereas the raw closest distance (d_c , Hyperforin closer) and the per-target influence (E , Hyperforin higher) rankings are stable at both thresholds. This is the effect-size/evidence dissociation of the main text, shown across network construction.

Metric	Compound	STRING ≥ 700		STRING ≥ 900	
		value	Z	value	Z
d_c (closest distance)	Hyperforin	0.60	−6.04	1.30	−3.86
	Quercetin	1.34	−5.46	1.68	−5.44
E (RWR influence)	Hyperforin	0.1212	+11.04	0.1138	+10.77
	Quercetin	0.0326	+5.46	0.0322	+4.58
E (EWI influence)	Hyperforin	0.1449	+10.35	0.1330	+9.28
	Quercetin	0.0511	+7.01	0.0493	+5.74

Table S7. Restart-probability (α) sensitivity (STRING ≥ 900). The Hyperforin > Quercetin per-target-efficiency ranking holds at every α ; the ratio is α -dependent and is not interpreted as an absolute constant.

α	0.10	0.15	0.20	0.30	0.50	0.70
E Hyperforin	0.0907	0.1138	0.1344	0.1716	0.2390	0.3032
E Quercetin	0.0312	0.0322	0.0322	0.0310	0.0271	0.0227
ratio	2.90	3.54	4.18	5.53	8.81	13.35

Table S8. Expression-floor sensitivity (EWI) (STRING ≥ 900). The 0.01 floor is immaterial; the ratio moves only in the third decimal.

floor	0	10^{-3}	10^{-2}	5×10^{-2}	10^{-1}
E Hyperforin	0.1330	0.1330	0.1330	0.1330	0.1327
E Quercetin	0.0493	0.0493	0.0493	0.0493	0.0491
ratio	2.695	2.695	2.695	2.695	2.702

Table S9. Direct-propagated decomposition of per-target influence: full controls (STRING ≥ 900). Four of ten Hyperforin targets are DILI-module members vs one of 62 Quercetin targets; isolating the propagated component (leave-one-out) and removing each source of direct overlap attenuates the per-target advantage to a modest residual.

Control	$E(\text{Hyp})$	$E(\text{Quer})$	ratio
Baseline (includes overlap)	0.1138	0.0322	3.54
Self-mass excluded	0.0427	0.0290	1.47
DILI-member seeds removed	0.0431	0.0295	1.46
CYP/transporter seeds removed	0.0338	0.0290	1.17
Shared targets removed (both)	0.0541	0.0288	1.88
<i>Propagated (leave-one-out) reference distributions ($n = 1,000$; Fig. 6B):</i>			
Global degree-matched random set ($n = 10$)	Hyp 0.0427 vs null mean 0.0130 (95th 0.0281; max 0.0580); 99.9th percentile, $3.3\times$ mean, $p = 0.002$, does not exceed all		
Size-matched Quercetin 10-subsets	Hyp 0.0427 vs subset mean 0.0296 (max 0.0622); $1.4\times$, overlapping		

Table S10. Network separation between target set and DILI module. Menche et al. separation measure $S_{AB} = \langle d_{AB} \rangle - (\langle d_{AA} \rangle + \langle d_{BB} \rangle)/2$ [4] (STRING ≥ 900); $S_{AB} > 0$ indicates topological separation. Both target sets are separated from the DILI module, and Hyperforin is *more* separated despite its four direct-overlap targets, because its targets form a tighter cluster ($\langle d_{AA} \rangle = 1.20$ vs 1.55). This corroborates the decomposition (Table 3, Fig. 6): Hyperforin’s per-target influence advantage arises from a few targets that *are* DILI genes (direct overlap), not from its target set being embedded in the DILI neighbourhood.

Target set	$\langle d_{AB} \rangle$	$\langle d_{AA} \rangle$	$\langle d_{BB} \rangle$	S_{AB}
Hyperforin	1.699	1.200	1.500	0.349
Quercetin	1.644	1.548	1.500	0.119

Table S11. Direct (distance-1) DILI connectivity of Hyperforin targets (regenerated on the STRING ≥ 900 liver LCC). Connectivity is concentrated in the cytochrome-P450 axis; the transporters (ABCB1, ABCC2, ABCG2) and the receptor NR1I2 have no additional distance-1 DILI neighbours (NR1I2 and ABCB1 are themselves module members, at distance 0). Per-target means: **Hyperforin 3.40** direct DILI neighbours (34 connections across 10 targets) vs **Quercetin 1.53** (95 across 62) — a $2.2\times$ enrichment that complements the distance-0 direct-overlap decomposition (Table S9).

Hyperforin target	n	Direct DILI neighbours (distance 1)
CYP3A4	11	CYP2A6, CYP2C19, CYP2C9, CYP2E1, FMO3, GSTM1, GSTM2, GSTP1, NR1I2, NR1I3, UGT1A9
CYP2C9	9	CYP2A6, CYP2C19, CYP2E1, FMO3, GSTM1, GSTM2, GSTP1, PTGS2, UGT1A9
CYP2B6	6	CYP2A6, CYP2C19, CYP2C9, CYP2E1, PTGS2, UGT1A9
AKT1	4	CTNNB1, IGF1, MAP3K5, NFE2L2
MMP9	3	LCN2, MMP2, SPP1
MMP2	1	LCN2
ABCB1, ABCC2, ABCG2, NR1I2	0	—

Table S12. DILI-module sensitivity: removing the entangled pharmacogenes. Every other robustness check in this work varies the *method* (α , expression floor, network threshold, degree-binning, two nulls); this one varies the *disease module*, the locus of the target–disease circularity. We remove from the 82-gene module the four genes that are simultaneously Hyperforin targets and DILI members (ABCB1, CYP2C9, MMP2, NR1I2), giving a reduced module D' that the drug targets cannot directly overlap, and recompute the propagated (leave-one-out) per-target influence symmetrically for both compounds (STRING ≥ 900 , $\alpha = 0.15$). The propagated advantage does *not* collapse — it slightly increases, because Hyperforin’s leave-one-out value is unchanged (those genes were already excluded from its own scoring) while Quercetin loses propagation onto ABCB1/CYP2C9/NR1I2, which it had benefited from as non-target DILI genes. Against a degree-matched random 10-gene background scored on D' ($n = 1,000$), Hyperforin remains at the 99th percentile ($3.45\times$ the background mean, empirical $p = 0.005$) but does not exceed every random set. The modest propagated signal supports the interpretation that the propagated residual is not eliminated after removing the overlapping pharmacogenes from the module.

Disease module	$ D $	Propagated E (Hyp)	Propagated E (Quer)	Ratio
Full DILI module	82	0.0427	0.0290	1.47
Overlap genes removed (D')	78	0.0427	0.0271	1.58

Table S13. Independent reimplementations of the canonical proximity toolbox: full null parameters (emreg00/toolbox, re-derived by GUNNEY_FIDELITY_check.py; STRING ≥ 900 ; $n = 1,000$; seed 42; $|D| = 82$ preserved). This is the detailed companion to the compact main-text summary (Table 4): for each null construction it reports the observed closest distance d_c , the null mean μ_{null} and standard deviation σ_{null} , and the resulting Z . The observed d_c reproduces the toolbox exactly (Hyperforin 1.300, Quercetin 1.677). The reimplementations’ $\pm 25\%$ window (group 1, Quercetin $Z = -5.35$) matches the committed `src` pipeline ($Z = -5.44$; Table S3) within permutation tolerance, confirming the two implementations agree. The $\pm 25\%$ window never silently falls back to unmatched nodes: every Hyperforin target retains ≥ 143 degree-matched candidates (its highest-degree target, AKT1, has degree 128). The evidence dissociation is clear under the fixed-disease null and attenuates to near-parity under the two-sided null; Hyperforin is the topologically closer compound throughout.

Null model	Compound	d_c	μ_{null}	σ_{null}	Z
$\pm 25\%$ window, fixed disease	Hyperforin	1.30	2.22	0.238	−3.86
	Quercetin	1.68	2.18	0.093	−5.35
Guney ≥ 100 -bin, fixed disease	Hyperforin	1.30	2.23	0.228	−4.09
	Quercetin	1.68	2.16	0.090	−5.34
Guney ≥ 100 -bin, two-sided	Hyperforin	1.30	2.05	0.212	−3.55
	Quercetin	1.68	2.01	0.090	−3.66

σ_{null} ratio (Hyperforin/Quercetin), from full-precision values: 2.56 ($\pm 25\%$ fixed), 2.53 (≥ 100 -bin fixed), 2.37 (≥ 100 -bin two-sided); expected $\sqrt{62/10} = 2.49$. The Law-of-Large-Numbers shrinkage therefore persists under every null construction, including Guney’s exact two-sided null.

Table S14. Quercetin target genes in the liver-expressed LCC (STRING ≥ 900 , GTEx v8 liver TPM ≥ 1 ; ChEMBL CHEMBL159). Sorted by descending TPM.

Gene	TPM	Gene	TPM	Gene	TPM	Gene	TPM
CFB	1115	CYP3A4	335	FN1	229	ALDH2	183
ANPEP	160	PPIA	112	SERPINA5	104	CYP1A2	72
CA2	64	APP	63	PYGL	55	HDAC6	45
ESRRA	42	MAOA	35	AKR1C2	33	AKT1	33
CTSH	28	XDH	26	CHRNA4	25	PIK3R1	24
PIM1	24	LDLR	23	ELOVL1	18	EGFR	17
PKN1	16	GSK3A	13	YES1	13	MET	12
DAPK1	12	BACE1	11	CSNK2A1	10	FSTL1	9
SIRT6	8	GSK3B	7	CDK7	7	CAV2	7
PTPN2	6	CYP1A1	5	PRMT7	5	MMP2	5
AKR1B1	5	PDE6D	5	BRAF	4	PTK2	4
ABCG2	4	IQGAP1	4	ADRB2	3	KDR	3
SRC	3	ALOX5	3	CYP1B1	3	TLR4	3
NUAK1	3	AXL	2	ADA	2	LCK	2
ABCC1	2	PLK1	1	ACHE	1	MMP9	1
SYK	1	PDZK1IP1	1				

Table S15. Curation provenance (stage counts, reproducible from the committed data). Targets are filtered to the intersection of the ≥ 700 and ≥ 900 liver LCCs, so target counts are identical at both thresholds; the DILI module is filtered per threshold.

Stage	Hyperforin	Quercetin	DILI module
Raw (lit. / ChEMBL / DisGeNET)	14	122	127
Processed (human, UniProt-mapped)	14	87	—
In ≥ 700 liver-expressed LCC	10	62	84
In ≥ 900 liver-expressed LCC	10	62	82

Table S16. DILI gene set (82 genes) in the liver-expressed STRING ≥ 900 LCC (DisGeNET, UMLS C0860207; GTEx v8 liver TPM ≥ 1), alphabetical. The four genes that are also Hyperforin targets (the distance-0 direct overlap, Table S9) are in bold.

82 DILI-associated genes							
ABCB1	AHR	ALB	ALDOB	AMBP	APOA1	APOE	APOH
ARG1	ARNT	ATG5	BAX	BTD	C3	CAT	CCL2
CLU	COL3A1	CTNNB1	CXCL1	CXCL10	CYP2A6	CYP2C19	CYP2C9
CYP2E1	DGAT2	ENO1	FGA	FLT1	FMO3	GADD45A	GC
GCLC	GPT	GSN	GSTM1	GSTM2	GSTP1	HLA-A	HLA-B
HLA-DQB1	HLA-DRB1	HMGB1	HMOX1	HPD	HPX	IGF1	IL18
IL1R2	KRT18	LCN2	LGALS3	MAP3K5	MED1	MMP2	MTHFR
NAT2	NFE2L2	NR1H3	NR1H4	NR1H2	NR1H3	PLAT	PLG
PNP	POLG	PON1	PPARA	PRKDC	PTGS2	RBP1	SLPI
SNX18	SOD1	SOD3	SPP1	TALDO1	TBXA2R	TCTN1	TF
TTR	UGT1A9						